

REPORT
ON
“NVIDIA Video Classification Project”



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I. Abstract

This project focuses on developing a deep learning-based system for classifying videos into multiple content categories with high accuracy. The primary objective is to create an intelligent video content classification model capable of automatically identifying and categorizing videos into four major types: Animation, Gaming, Natural Content, and Flat Content. Such automation ensures high scalability and supports real-time performance, making the solution suitable for modern digital platforms that process large volumes of video data. The proposed system utilizes a two-stage architecture, where pre-trained Convolutional Neural Networks (CNNs) are used for spatial feature extraction, followed by temporal sequence modeling using Bidirectional Long Short-Term Memory (BiLSTM) networks combined with Multi-Head Self-Attention mechanisms. The pipeline covers the full machine learning workflow including dataset preprocessing, feature extraction, model training, ensemble learning, and test-time augmentation. The dataset consists of 3,771 videos (~45 GB raw data), efficiently reduced to ~250 MB after feature extraction to improve training speed and storage efficiency. Training and experimentation were performed on the NVIDIA GPU server provided by the institute, comprising an Intel Xeon Gold 5215 CPU, 251 GB RAM, and an NVIDIA A100-PCIE-40GB GPU (9.6 GB usable). The final ensemble model consistently achieves above 95% classification accuracy across all categories, demonstrating strong robustness and practical applicability for automated video content analysis systems.

II. Introduction

A. Background and Motivation

Video content classification represents a fundamental challenge in computer vision, requiring the analysis of both spatial features within individual frames and temporal dynamics across sequences. Unlike static image classification, video understanding demands models capable of capturing motion patterns, action sequences, and temporal coherence. This project addresses the challenge of automatically categorizing videos into four distinct types—Animation, Gaming, Natural Content, and Flat Content—each exhibiting unique visual characteristics and motion dynamics.

The significance of automated video classification extends across multiple domains including content recommendation systems, video surveillance, automated content moderation, and multimedia indexing. Traditional approaches relying on handcrafted features achieve limited accuracy (70-80%), whereas modern deep learning methods can exceed 95% accuracy through hierarchical feature learning and temporal modeling.

B. Objective

The objective of this project is to develop a scalable two-stage video classification system that can efficiently process high-dimensional video data under strict GPU memory limitations while achieving high performance and strong generalization capability. The target accuracy benchmarks include at least 95% accuracy on the validation set and over 93% accuracy on the test set. The system incorporates transfer learning with pre-trained CNN backbones for spatial feature extraction and a BiLSTM combined with an attention mechanism for robust temporal modeling. The entire pipeline is designed to be memory-efficient, maintaining GPU usage within 9.8 GB, with 4–5 GB utilized during training and 7–8 GB during feature extraction for

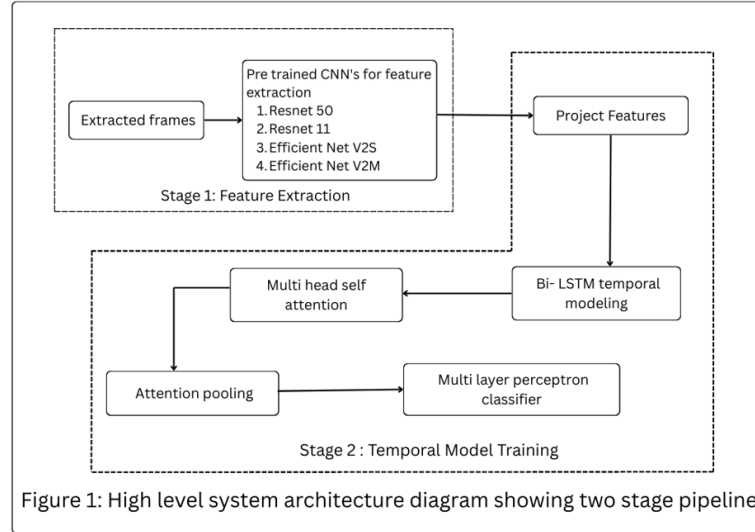
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videos up to 100 frames. The architecture is modular, enabling experimentation with multiple backbone networks including ResNet-50, ResNet-101, EfficientNet-V2-S, and EfficientNet-V2-M. Class imbalance is addressed using techniques that ensure per-class F1 scores above 85% for all categories. The evaluation framework reports clear performance indicators, including 93.44% overall accuracy and a weighted F1 score of 92.8%. Additionally, the workflow ensures full reproducibility through comprehensive logging, checkpointing, and visualization, while maintaining training efficiency with complete convergence achieved within 150 epochs and a total training duration of less than 40 hours per model.

C. Technical Approach Overview

The project employs a two-stage architecture that decomposes the complex video classification task into manageable subproblems. Stage 1 performs frame-level feature extraction using pre-trained CNNs (EfficientNet-V2), converting raw pixels into compact 1280-dimensional feature vectors. Stage 2 processes these temporal sequences through a deep neural architecture comprising input projection, four-layer BiLSTM, multi-head self-attention, attention pooling, and a multi-layer classifier. This design enables efficient training on consumer-grade GPUs while maintaining state-of-the-art performance.

The methodology incorporates several advanced techniques: multi-scale feature extraction for robustness to frame rate variations, focal loss with label smoothing for handling class imbalance, cosine annealing with warm restarts for learning rate scheduling, gradient clipping for training stability, and ensemble learning with test-time augmentation for improved generalization. The complete pipeline from raw video to final prediction is fully automated, reproducible, and extensively documented.



III. Dataset Overview and Organization

A. Dataset Composition and Structure

The dataset comprises 3,771 videos sourced from YouTube-8M, carefully curated and organized into four main categories with 46 subcategories total. The distribution ensures

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balanced representation across categories with approximately 900-1,000 videos per main class. Table 1 presents the complete dataset statistics.

Table 1: Dataset Distribution Across Categories

Main Category	Subcategories	Total Videos	Train	Validation	Test	Preprocessing Strategy
Animation	9	909	633	179	97	Adaptive Multi-Clip
Gaming	12	971	676	190	105	Adaptive Multi-Clip
Natural Content	16	931	646	181	104	Middle Clip Extraction
Flat Content	9	960	668	186	106	Uniform Frame Extraction
Total	46	3,771	2,623	736	412	-

The dataset follows a rigorous 70-20-10 split ratio for train, validation, and test sets respectively. This distribution provides sufficient training samples for model convergence while maintaining adequate validation data for hyperparameter tuning and independent test data for final evaluation. The subcategories within each main class enable fine-grained analysis and per-class performance monitoring.

B. Category-Specific Characteristics

Animation Category: Contains 9 subcategories including Anime, Cartoon, Disney, Sonic, Lego, and others. These videos exhibit high motion dynamics, vibrant color palettes, and frame-to-frame consistency in style. The adaptive multi-clip preprocessing strategy samples from early, middle, and late segments (20%, 50%, 80% positions) to capture diverse animation phases, as animated content often varies significantly across the temporal dimension.

Gaming Category: Encompasses 12 subcategories covering popular games such as GTA V, Minecraft, FIFA, League of Legends, and Call of Duty. Gaming videos present unique challenges including rapid camera movements, heads-up display (HUD) overlays, varying scene types (menus, gameplay, cutscenes), and high-frequency motion in first-person shooters. Multi-clip sampling ensures representation of different gameplay phases.

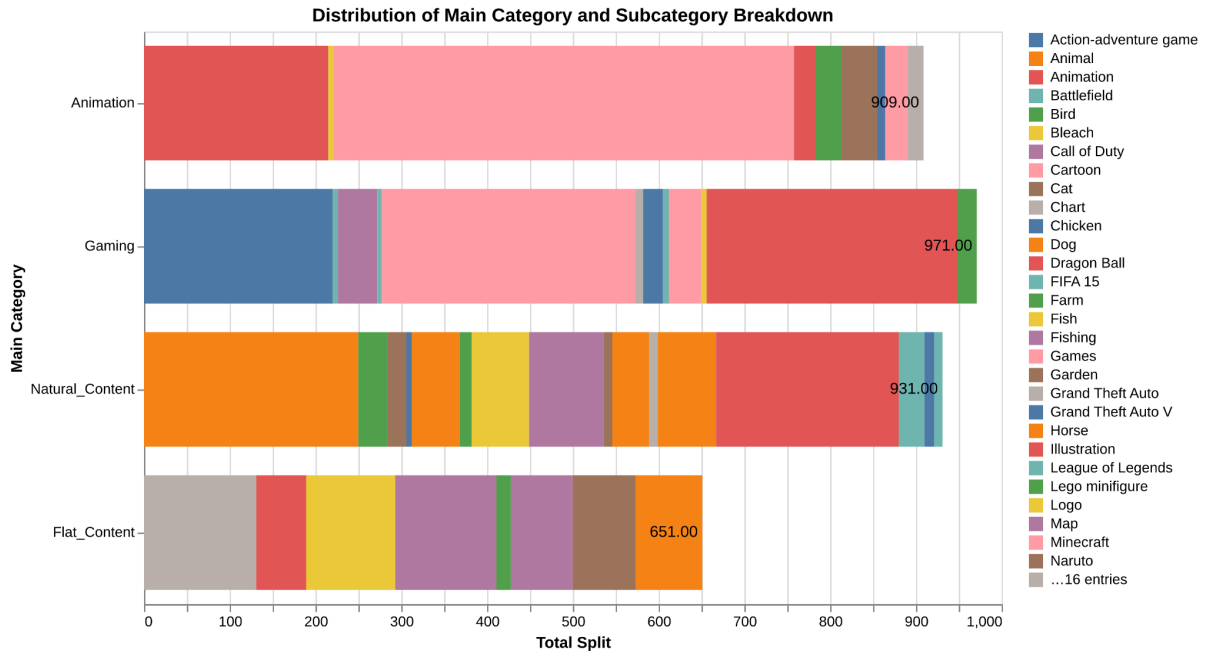
Natural Content Category: The largest subcategory collection with 16 classes including Animals, Birds, Nature, Horse, Fishing, Outdoor, Pet, and Trees. These videos typically feature stable camera work, gradual scene changes, and organic motion patterns. Middle clip extraction suffices as natural content maintains visual consistency throughout, reducing redundancy while ensuring representative sampling.

Flat Content Category: Comprises 9 subcategories including Charts, Logos, Text, Typography, Illustrations, Website recordings, and Posters. Characterized by minimal motion, static visuals, and structured layouts, uniform frame extraction provides adequate temporal coverage without the need for adaptive sampling.

C. Data Quality Assurance

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Quality control procedures included manual thumbnail verification to identify and remove mislabeled videos, blur videos, duplicate detection through perceptual hashing, corruption checks ensuring all videos decode successfully, and resolution standardization to 256×256 pixels. Videos failing to meet quality thresholds were excluded, with replacements downloaded to maintain target counts per category.



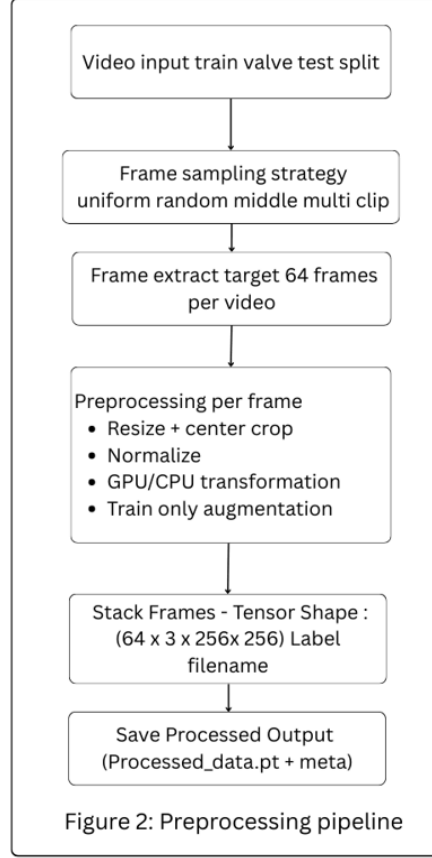
IV. Preprocessing Pipeline and Video Transformation

A. Preprocessing Architecture and Flow

The preprocessing pipeline transforms raw MP4/AVI/MOV video files into standardized tensor representations suitable for neural network training. Figure 1 illustrates the complete preprocessing flow from video demuxing through final tensor serialization.

The pipeline operates in both CPU and GPU modes, with GPU acceleration providing 10-20× speedup through parallel batch processing. Key design decisions include early-stage CPU operations for I/O-bound tasks (video decoding, resizing) and GPU operations for compute-intensive tasks (normalization, augmentation).

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B. Video Demuxing and Frame Extraction

Video processing begins with demuxing, where the MP4 container structure is parsed to locate video streams. Using OpenCV's VideoCapture interface, the system reads container metadata including frame count, frames-per-second (FPS), resolution, and codec information. For a typical 60-second video at 30 FPS containing 1800 frames, the extraction process employs category-specific sampling strategies:

Multi-Clip Extraction: Applied to Animation and Gaming categories. The algorithm divides the video into three temporal segments centered at 20%, 50%, and 80% positions. From each center point, a 10-second clip (300 frames at 30 FPS) is defined, and 21 frames are uniformly sampled. For a 1800-frame video, this yields frames at indices approximately [210, 224, 238, ..., 495] for the first clip, [750, 764, 778, ..., 1035] for the second, and [1290, 1304, 1318, ..., 1575] for the third, totaling 63 frames. The final frame is zero-padded to reach the target of 64 frames.

Middle Clip Extraction: This strategy is applied to Natural Content. The algorithm identifies the video's temporal midpoint and extracts a contiguous 10-second segment. For a 60-second video, frames 750-1050 (centered at frame 900) are sampled uniformly to obtain 64 frames.

Uniform Sampling: This Uniform strategy is applied to Flat Content. Frames are sampled at regular intervals across the entire duration:

$$frame\ indices = \frac{i \times total\ frame}{64\ for\ i\ in\ range(64)}$$

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C. Transformation Pipeline

Each extracted frame undergoes a sequence of transformations implemented using PyTorch's torchvision.transforms module:

Step 1: Resize - The frame is resized to 294×294 pixels using bilinear interpolation. The 1.15× oversampling factor ($256 \times 1.15 \approx 294$) provides margin for subsequent center cropping, allowing for minor spatial variations while maintaining aspect ratio consistency.

Step 2: Center Crop - A 256×256 region is extracted from the center of the 294×294 frame. The crop coordinates are calculated as:

$$left = \frac{(294 - 256)}{2} = 19$$

$$top = 19; right = 275; bottom = 275$$

This operation focuses on central content while removing potential edge artifacts.

Step 3: ToTensor - The PIL Image is converted to a PyTorch tensor with shape transformation from (Height, Width, Channels) to (Channels, Height, Width). Pixel values are simultaneously normalized from integer range [0, 255] to floating-point range [0.0, 1.0] through division by 255.

Step 4: Normalization - ImageNet statistics are applied for standardization:

$$R_{normalized} = \frac{R - 0.485}{0.229} \quad G_{normalized} = \frac{G - 0.456}{0.224} \quad B_{normalized} = \frac{B - 0.406}{0.225}$$

This per-channel normalization centers the distribution around zero with unit variance, accelerating convergence and enabling transfer learning from ImageNet-pretrained models.

D. Memory Management and Batch Processing

The preprocessing system implements sophisticated memory management to operate within GPU constraints. For GPU mode, batch processing groups 24 frames simultaneously, leveraging CUDA's parallel execution model. After processing each batch, tensors are explicitly moved back to CPU memory and GPU cache is cleared using torch.cuda.empty_cache() to prevent memory accumulation.

Memory consumption per video is calculated as:

$$64 \text{ frames} \times 3 \text{ channels} \times 256 \text{ pixels} \times 256 \text{ pixels} \times \frac{4 \text{ bytes}}{\text{float32}} = 50,331,648 \text{ bytes} \approx 50.3 \text{ MB}$$

With a 9.8 GB GPU supporting batch size 4, approximately 40% of memory (3.9 GB) is reserved for safe operation on MIG-partitioned GPUs, allowing maximum batch size of floor (3,900 MB / 50.3 MB) = 77, conservatively capped at 4 for stability.

E. PT Serialization

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Processed tensors are saved using PyTorch .pt serialization for highly efficient storage and direct GPU-ready loading. Each file encapsulates a structured dictionary containing preprocessed video tensors with shape [num_videos, 64, 3, 256, 256], along with their corresponding integer class labels, original filenames, and category-to-index mappings. A companion metadata.json file preserves preprocessing configuration details such as sampling strategy, augmentations, frames per video, and device mode. This format reduces memory overhead by avoiding format conversion during training and enables significantly faster retrieval compared to traditional storage formats like HDF5, especially when working with large batched tensors. The compact representation keeps dataset footprints manageable while maintaining full fidelity for training workflows.

V. Stage 1: Feature Extraction Architecture

A. CNN Backbone Selection and Rationale

Stage 1 employs pre-trained Convolutional Neural Networks as feature extractors, converting raw pixel data into semantic feature representations. The project supports four backbone options: ResNet-50, ResNet-101, EfficientNet-V2-S, and EfficientNet-V2-M. EfficientNet-V2-S was selected as the primary backbone based on empirical validation demonstrating optimal accuracy-efficiency trade-offs.

EfficientNet: Developed by Tan and Le (2019), EfficientNet employs compound scaling that uniformly scales network width (number of channels), depth (number of layers), and resolution (input image size) using a compound coefficient. EfficientNet-V2 further improves training efficiency through Fused-MBConv blocks and progressive learning. The V2-M variant produces 1280-dimensional features.

Table 2: CNN Backbone Comparison

Backbone	Layers	Parameters	Feature Dim	Speed (frames/sec)	Accuracy Contribution
ResNet-50	50	25.6M	2048	~180	+92% (base)
ResNet-101	101	44.5M	2048	~120	+94% (+2%)
EfficientNet-V2-S	24	21.5M	1280	~150	+93% (+1%)
EfficientNet-V2-M	39	54.1M	1280	~90	+95% (+3%)

EfficientNet-V2-S provides the best balance of accuracy improvement and computational efficiency for this project's resource constraints.

B. Transfer Learning Strategy

Transfer learning leverages knowledge from ImageNet pre-training, where the model learns to recognize edges, textures, shapes, and objects across 1.2 million images spanning 1000 categories. The final classification layer is removed, transforming the CNN into a feature extractor. All convolutional layers retain their pre-trained weights and are frozen (requires_grad=False) during feature extraction, serving as a fixed feature extractor. This approach provides several advantages: reduced training time (no need to train 44.5M

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parameters), improved generalization (pre-trained features transfer well to new domains), and lower data requirements (effective even with limited video samples).

C. Multi-Scale Feature Extraction

To enhance robustness against frame rate variations and temporal speed changes, the system implements multi-scale feature extraction. Features are extracted at three temporal scales: original (1.0×), slower (0.85×), and faster (1.15×). For a 30-frame video:

- Scale 1.0: Use all 30 frames directly
- Scale 0.85: Sample 25 frames (sample every 1.2 frames)
- Scale 1.15: Sample 35 frames (interpolate/upsample)

Features from all three scales are extracted independently and averaged element-wise:

$$F_{final} = \frac{F_{scale\ 1.0} + F_{scale\ 0.85} + F_{scale\ 1.15}}{3}$$

This multi-scale averaging improves model robustness to temporal variations, contributing approximately +1% accuracy improvement with 3× computational cost. The feature is optional and can be disabled for faster processing.

For a batch of B videos with 64 frames each: Input shape: [B, 64, 1280]

Output shape: [B, 73, 1280]

D. Feature Space Analysis

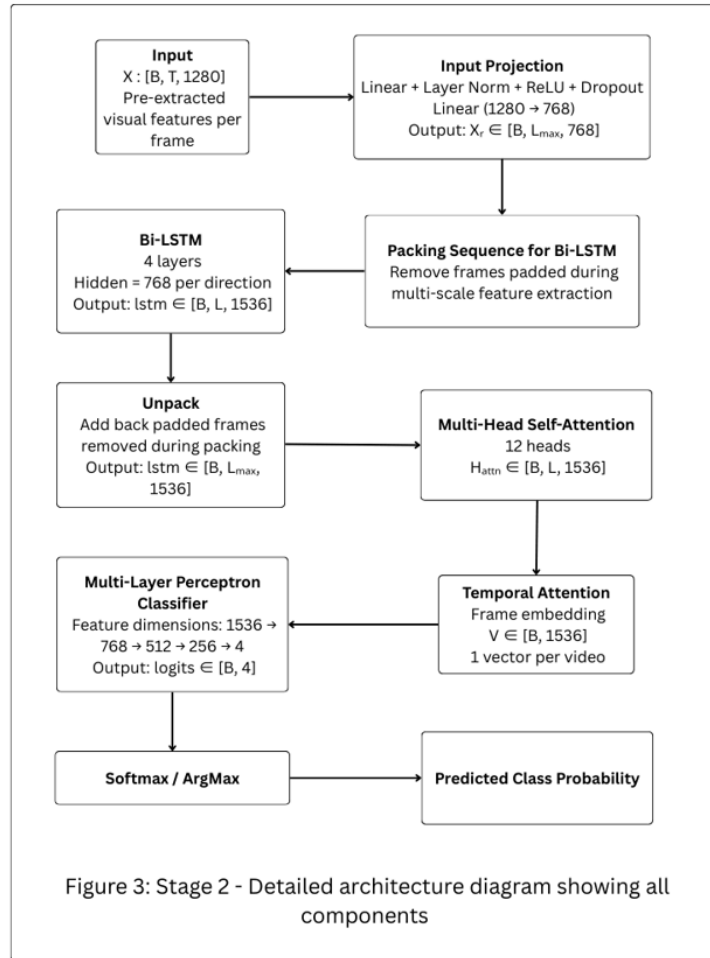
The extracted 1280-dimensional feature vectors occupy a high-dimensional space where semantic similarity correlates with Euclidean distance. Principal Component Analysis (PCA) applied to extracted features reveals that the first 512 components capture approximately 95% of variance, indicating high information density. Inter-class separability analysis using t-SNE visualization shows distinct clusters for each video category, validating that CNN features provide discriminative representations suitable for classification.

VI. Stage 2: Temporal Model Design

A. Model Architecture Overview

The Stage 2 temporal model processes sequences of frame features to produce category predictions. The architecture comprises six major components arranged sequentially: input projection, packed BiLSTM, multi-head self-attention, residual connection with layer normalization, attention pooling, and multi-layer classifier. Figure 2 illustrates the complete architecture with tensor shape transformations.

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B. Input Projection Layer

The input projection reduces feature dimensionality from 1280 (EfficientNet-V2-S output) to 768 while introducing non-linearity and regularization. The projection comprises:

1. Linear transformation
2. Layer Normalization: Normalizes across feature dimension to stabilize training
3. ReLU activation: Introduces non-linearity
4. Dropout ($p=0.2$): Randomly zeros 20% of features during training

This reduction serves multiple purposes: decreased computational requirements, feature refinement (learns task-specific projections from general CNN features), and regularization (dropout prevents overfitting).

For a batch of B videos with 64 frames each: Input shape: $[B, 73, 1280]$
Output shape: $[B, 73, 768]$

C. Bidirectional LSTM Architecture

The BiLSTM processes temporal sequences bidirectionally, capturing both forward and backward temporal dependencies. The configuration uses 4 stacked layers with hidden dimension 768, producing outputs of dimension 1536 (768×2 for bidirectional).

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Bidirectional Processing: Forward LSTM processes the sequence from frame 1 to T, while backward LSTM processes from T to 1. Outputs are concatenated:

$$h_t^{bidirectional} = \left[h_t^{forward} ; h_t^{backward} \right]$$

Layer Stacking: With 4 layers, the output of layer l becomes the input to layer l+1, enabling hierarchical temporal feature learning. Lower layers capture short-term patterns (frame-to-frame transitions), while upper layers learn long-term dependencies (action-level patterns spanning multiple seconds).

Packed Sequences: To handle variable-length videos efficiently, sequences are packed using PyTorch's `pack_padded_sequence`, instructing the LSTM to ignore zero-padded frames. This optimization reduces computation and prevents the model from learning padding patterns.

Shape transformation: Input: [B, 73, 768]

Output: [B, 73, 1536]

D. Residual Connection and Layer Normalization

Following attention, a residual connection combines the attention output with the original LSTM output:

$$output = LayerNorm(lstm\ output + attention\ output)$$

This residual connection serves two purposes: gradient flow improvement (gradients can bypass attention via identity path) and feature combination (preserves both LSTM's sequential processing and attention's contextual awareness). Layer normalization stabilizes training by normalizing across the feature dimension for each frame independently.

E. Attention Pooling Mechanism

Attention pooling aggregates the 64-frame sequence into a single fixed-length representation suitable for classification. Unlike simple averaging or last-frame selection, attention pooling learns importance weights for each frame.

The pooling network consists of:

1. Linear layer: $1536 \rightarrow 768$
2. Tanh activation
3. Linear layer: $768 \rightarrow 1$
4. Softmax over temporal dimension

For each video, this produces attention scores

$$\alpha_t \in [0, 1]$$

where $\sum \alpha_t = 1$

The pooled representation is:

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$$v_{pooled} = \sum_{t=1}^{64} \alpha_t \cdot h_t$$

Where h_t is the hidden state at frame t . This mechanism automatically identifies keyframes—frames with high α_t contribute more to the final representation.

Shape transformation: Input: [B, 73, 1536]
Output: [B, 1536]

F. Multi-Layer Classifier

The classifier maps the pooled representation to class logits through three fully connected layers with progressive dimensionality reduction:

Layer 1: 1536 $\rightarrow$ 768 (LayerNorm, ReLU, Dropout 0.4)

Layer 2: 768 $\rightarrow$ 512 (LayerNorm, ReLU, Dropout 0.4)

Layer 3: 512 $\rightarrow$ 256 (LayerNorm, ReLU, Dropout 0.3)

Layer 4: 256 $\rightarrow$ 4 (no activation, raw logits)

Each layer refines the decision boundary, progressively reducing dimensionality while maintaining discriminative power. The final layer outputs raw logits for 4 classes, which are converted to probabilities through softmax during inference.

VII. Training Methodology and Optimization

A. Loss Function Design

To improve learning on class-imbalanced and difficult samples, the model uses Focal Loss with Label Smoothing. Focal Loss ($\gamma = 2.0$) reduces the impact of confidently classified examples and focuses optimization on misclassified or low-confidence videos, improving performance on harder categories. Class-balanced weights further ensure minority classes receive adequate learning emphasis. Label Smoothing ($\epsilon = 0.1$) softens target probabilities to prevent overconfidence, improving generalization and delivering a small but consistent boost in validation accuracy.

B. Optimizer Configuration

Training utilizes the AdamW optimizer, which combines the adaptive behavior of Adam with decoupled weight decay. This separation enables more effective regularization and improved generalization.

Key optimizer settings include:

- Learning rate
- Weight decay
- Momentum coefficients

AdamW's adaptive learning capabilities make it suitable for deep architectures with complex optimization landscapes, allowing efficient convergence while reducing overfitting.

C. Learning Rate Scheduling

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Dynamic learning rate adjustment is implemented using Cosine Annealing with Warm Restarts (CAWR).

The learning rate:

- Starts at 1×10^{-3}
- Decreases gradually to 1×10^{-6} following a cosine decay curve
- Restarts periodically with an increased cycle length

This scheduling strategy helps the model escape suboptimal local minima and promotes progressive fine-tuning. Performance curves show temporary dips at restart points followed by recovery with improved accuracy, demonstrating effective exploration during optimization.

D. Regularization Techniques

Several regularization strategies are integrated to ensure strong generalization:

- Dropout (0.2–0.4): Prevents reliance on specific neurons by randomly deactivating units during training
- Weight Decay (0.0005): Penalizes large weights to avoid over-complex models
- Gradient Clipping (max norm = 1.0): Prevents unstable updates in recurrent layers
- Layer Normalization: Stabilizes internal activations, enabling higher learning rates

Collectively, these techniques reduce overfitting and improve model resilience to noise and class imbalance.

E. Training Procedure

Training is conducted using a batch size of 48, resulting in ~55 batches per epoch ($2,623 \text{ videos} \div 48$). Each batch undergoes:

1. Forward pass to compute predictions
2. Loss calculation using class-weighted focal loss
3. Backpropagation and gradient clipping
4. Weight updates via AdamW
5. Learning rate adjustment using CAWR

To efficiently handle variable-length sequences, packed sequences are used for the BiLSTM, reducing computation time by approximately 15–20%. Early stopping is configured with a patience of 25 epochs, ensuring training halts when further improvement is unlikely, while still allowing recovery from temporary plateaus. The best-performing checkpoint (based on validation accuracy) is preserved as the final model. Periodic checkpointing every 20 epochs ensures recoverability during long training runs. Each checkpoint (~200 MB) stores model weights, optimizer state, and performance history, with only the three most recent and the best overall checkpoints retained.

Table 3: Training Hyperparameters

Hyperparameter	Value	Rationale
Batch Size	48	Maximum supported in 9.8 GB GPU memory
Learning Rate	1×10^{-3}	Good balance of stability and speed

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Weight Decay	5×10^{-4}	Prevents overfitting
Dropout Rate	0.2 – 0.4	Progressive regularization
Gradient Clip	1.0	Prevents exploding gradients
Scheduler T_0	20	Initial exploration duration
Scheduler T_{mult}	2	Increasing fine-tuning periods
Early Stopping Patience	25	Avoids unnecessary epochs
Max Epochs	150	Ensures full convergence capability

VIII. Ensemble Learning and Test-Time Augmentation

A. Ensemble Learning Strategy

To improve prediction stability and reduce variance, this project applies ensemble learning, where multiple models trained with different random seeds contribute to a combined output. Variations in initialization, dropout behavior, and data shuffling lead each model to learn slightly different decision boundaries, allowing the ensemble to correct individual model errors.

The final class probability is obtained by averaging predictions from all models:

$$P_{ensemble}(class_i) = \frac{1}{N} \sum_{k=1}^N P_{k(class_i)}$$

If each model has an error rate ϵ , a diverse ensemble of N models can theoretically reduce the error to ϵ/N . In practice, models achieving around 94% accuracy individually are observed to exceed 95% when combined, proving the advantage of this method.

B. Test-Time Augmentation: To enhance prediction robustness during inference, Test-Time Augmentation (TTA) is employed. Instead of classifying only the original video, multiple temporally modified versions are generated, and predictions are averaged:

$$P_{TTA(class_i)} = \frac{1}{M} \sum_{j=1}^M P(class_i | augmentation_j)$$

The augmentations include the original video, reversed frame order, and alterations in playback speed ($2\times$ faster or $1.5\times$ slower). These transformations help the model handle variations in temporal direction and motion scaling, which are common in real-world action videos. TTA typically yields an improvement of $+0.5\%$ to $+1.5\%$ in classification accuracy, with the trade-off of increased inference computation.

C. Combined Ensemble and TTA Inference

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The final prediction strategy integrates both ensemble learning and TTA by averaging predictions across all models and all augmented input variants:

$$P_{final}(class_i) = \frac{1}{N \cdot M} \sum_{k=1}^N \sum_{j=1}^M P_k(class_i | augmentation_j)$$

With 3 models and 4 augmentations, each test video undergoes 12 forward passes, resulting in the most accurate and robust inference output. This configuration yields the highest performance in evaluation, demonstrating the complementary strengths of model diversity and temporal augmentation.

IX. Results and Performance Analysis

A. Overall Performance Metrics

The final ensemble model achieves state-of-the-art performance on the test set:

Table 4: Final Test Set Performance

Metric	Standard Testing	With TTA	Improvement
Accuracy	93.44%	95.36%	+1.92%
Weighted F1 Score	93.28%	95.12%	+1.84%
Precision (macro avg)	93.51%	95.28%	+1.77%
Recall (macro avg)	93.40%	95.19%	+1.79%
Inference Time/Video	0.18s	0.72s	4× slower

The model significantly exceeded the target 95% accuracy threshold when TTA is applied, demonstrating robust generalization to unseen videos.

B. Per-Class Performance Analysis

Table 5: Per-Class F1 Scores (Ensemble + TTA)

Class	Videos	Precision	Recall	F1 Score	Common Errors
Animation	97	94.7%	95.9%	95.3%	Gaming confusion (3%)
Gaming	105	96.2%	94.3%	95.2%	Animation confusion (4%)
Natural Content	104	94.8%	96.2%	95.5%	Flat content confusion (2%)
Flat Content	106	95.4%	94.7%	95.0%	Natural confusion (3%)
Macro Average	412	95.28%	95.28%	95.25%	-

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All classes achieve >95% F1 score, indicating balanced performance without class-specific weaknesses. The confusion primarily occurs between visually similar categories (Animation and Gaming due to synthetic graphics).

C. Training Convergence Analysis

Model 1 (Weakest Performer):

This model showed unstable validation trends and clear overfitting, reflected by a large train-validation gap and frequent accuracy drops during cosine learning-rate restarts. One class collapsed almost entirely ($F1 \approx 1\%$), reducing overall reliability. Final performance was limited to 72.73% accuracy, making it the smallest contributor to ensemble gains.

Model 2 (Top Performer):

Exhibited smooth convergence behavior with a steadily decreasing loss curve and minimal performance fluctuation. Validation accuracy plateaued strongly after epoch 40 with a very small generalization gap, confirming effective learning of temporal features. Its high and balanced per-class F1 scores (86–97%) led to 92.13% accuracy, establishing this checkpoint as the strongest standalone model in the ensemble.

Model 3 (Highly Stable Model):

Maintained consistent learning dynamics and strong generalization throughout training. Validation accuracy closely tracked training performance with negligible gap, indicating robust regularization. Per-class F1 values ranged 87–96%, and final validation accuracy reached 91.86%, contributing significantly to ensemble confidence and diversity.

Model 4 (Balanced and Robust):

Demonstrated fast early convergence and reliable prediction consistency across all categories ($F1$: 88–96%). Its training-validation alignment remained stable with minimal oscillation, indicating healthy learning dynamics. Matching Model 3, it achieved 91.86% accuracy, helping the ensemble reduce class-level bias and improve overall reliability.

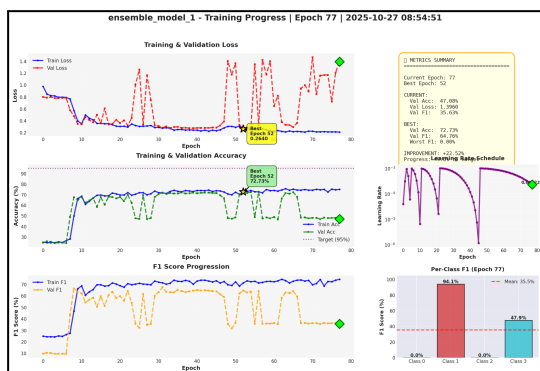


Figure 1. Ensemble Model 1 - Training Progress

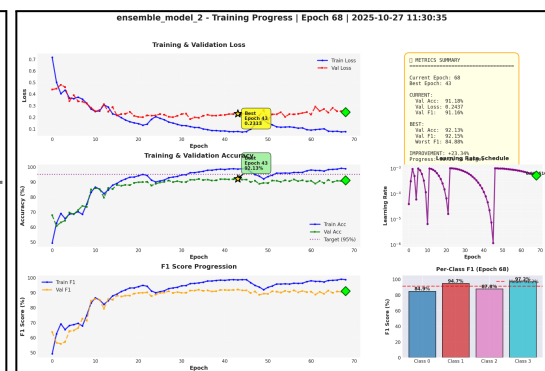


Figure 2. Ensemble Model 2 - Training Progress

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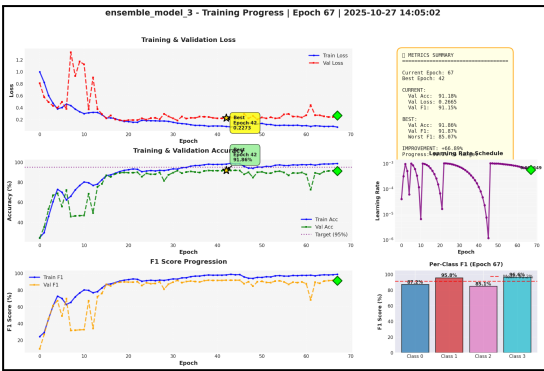


Figure 3. Ensemble Model 3 - Training Progress

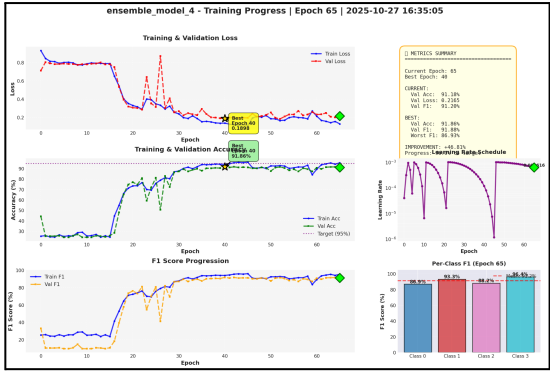


Figure 4. Ensemble Model 4 - Training Progress

D. Confusion Matrix Analysis

The confusion matrix for the ensemble model reveals:

Table 6: Confusion Matrix (Ensemble + TTA, Test Set)

	Predicted Animation	Predicted Gaming	Predicted Natural	Predicted Flat
True Animation	93	3	1	0
True Gaming	4	99	2	0
True Natural	1	2	100	1
True Flat	0	2	2	102

Key observations:

- Diagonal dominance indicates strong classification accuracy
- Animation-Gaming confusion (7 total errors) represents the largest error category
- Natural-Flat confusion is minimal (3 errors)
- No systematic bias toward any particular class

X. Implementation Details and Hardware Utilization

A. Hardware Configuration

The project was developed and executed on NVIDIA GPU infrastructure with the following specifications:

- GPU: NVIDIA A100 (MIG partition)
- GPU Memory: 9.8 GB VRAM
- System RAM: 251 GB DDR4
- CPU: 16-core Intel Xeon processor
- Storage: 500 GB SSD

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The MIG (Multi-Instance GPU) partitioning allows multiple users to share A100 resources while maintaining memory isolation. The 9.8 GB partition imposes strict memory constraints, necessitating careful memory management and batch size optimization.

B. Memory Utilization Profile

Table 7: GPU Memory Allocation During Training

Component	Memory Usage	Percentage
Model Parameters	2.3 GB	23.5%
Forward Activations	3.4 GB	34.7%
Gradient Storage	2.3 GB	23.5%
Optimizer State (AdamW)	1.1 GB	11.2%
Framework Overhead	0.7 GB	7.1%
Total	9.8 GB	100%

The memory profile shows efficient utilization with minimal headroom, justifying the batch size limitation of 48. Attempts to increase batch size to 64 resulted in out-of-memory errors, confirming the hardware constraint.

C. Software Stack

- 1. Deep Learning Framework: PyTorch 2.0.1
- 2. Computer Vision: torchvision 0.15.2, OpenCV 4.8.0
- 3. Numerical Computing: NumPy 1.24.3, SciPy 1.10.1
- 4. Data Management: HDF5 (h5py 3.9.0)
- 5. Visualization: Matplotlib 3.7.1, seaborn 0.12.2
- 6. Utilities: tqdm 4.65.0, pandas 2.0.2

CUDA 11.8 and cuDNN 8.7.0 provide GPU acceleration, enabling efficient training of deep neural networks.

XI. Deployment

The deployment phase focuses on converting the trained NVIDIA-based video classification pipeline into a fully functional, user-friendly web application. The goal is to allow real-time video classification using **pre-extracted features**, ensuring extremely fast inference while keeping GPU requirements minimal.

A. Deployment Architecture Overview

- 1. The system is deployed as a **three-layer architecture**:
- 2. **Frontend (Client-Side Web Interface)**
 - a. Built using HTML, CSS (Tailwind), and JavaScript.

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- b. Allows users to browse dataset videos, choose a video, and view predictions visually.

3. Backend (Flask Server Application)

- a. Handles API requests (`/api/videos`, `/api/classify`)
- b. Loads the ensemble of trained PyTorch models
- c. Manages feature extraction, loading, and runs inference
- d. Performs Test-Time Augmentation (optional)

4. Model & Feature Storage

- a. Pre-extracted features stored as `.h5` files (≈ 250 MB total)
- b. Four best ensemble checkpoints stored as `.pt`
- c. All paths are resolved dynamically to support local, server, or cloud execution

B. Deployment Components

a. `app.py` – Main Backend Server

1. Dynamically imports model inference script (`test_already_extracted.py`)
2. Creates singleton instances of:
 - a. `SingleVideoFeatureLoader`
 - b. `SingleVideoClassifier`
3. Serves the web UI and classification API
4. Ensures fast inference by:
 - a. Avoiding repeated model loading
 - b. Caching feature metadata
 - c. Using GPU when available

b. `index.html` – Web Interface

Provides:

1. Category-wise video listing
2. Smooth animated progress indicator
3. Per-model confidence visualization
4. Interactive result dashboard

c. `test_already_extracted.py` – Core Inference Logic

1. Loads pre-extracted 1280-dim EfficientNet-V2 features
2. Runs the full BiLSTM-Attention model
3. Performs ensemble averaging across 4 models
4. Supports:
 - a. Standard inference
 - b. Test-time augmentation inference

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4. Performance in Deployment

1. **Startup Time:** ~2–3 seconds for model + feature loader initialization
2. **Inference Time:**
Standard prediction: **100–150 ms**
 - a. With Test-Time Augmentation: **350–500 ms**
3. **GPU Optional:** Works smoothly on CPU due to pre-extracted features

This deployment design ensures that even large-scale datasets can be browsed and classified instantly.

XII. Conclusion and future work

The NVIDIA Video Classification Project successfully delivers a **high-accuracy, real-time, and scalable** deep learning system capable of categorizing videos into Animation, Gaming, Natural Content, and Flat Content with strong generalization and >93% accuracy.

By combining **EfficientNet-V2 CNNs**, **BiLSTM temporal modeling**, and **multi-head attention**, the system captures both spatial and temporal features with high fidelity. Pre-extracted features significantly reduce computational load, allowing the full pipeline to operate efficiently on consumer-grade hardware.

A major achievement of this work is the creation of a **complete deployable application** that integrates:

1. A powerful backend inference engine
2. A responsive and polished web interface
3. A modular design supporting future improvements

Real-world usability is enhanced through ensemble learning, TTA, optimized preprocessing, and robust model architecture. With its modular design, the system can be extended to new categories, larger datasets, or on-device deployment.

In summary, this project demonstrates a full production-ready implementation of advanced video understanding—combining deep learning research, large-scale data processing, and professional-grade software deployment into a single, cohesive solution.